

Solutions and Notes of Homework 1

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In linear algebra, the *concepts* are as important as the *computations*. The simple numerical exercises only help you check your understanding of basic procedures. Later in your career, **computers will do the calculations**, but you will have to **choose** the calculations, know how to **interpret** the results, and then **explain** the results to other people.

You must avoid the temptation to look at answers before you have tried to write out the solution yourself. **Otherwise, you are likely to think you understand something when in fact you do not.**

—*Linear Algebra and Its Applications*, by David C. Lay et al.

1. Justify each answer. Draw ✓ if True and × if False.

- (a) The empty set is not a vector space. (✓)

Explanation The empty set does not contain an additive identity (zero vector), which is a requirement for a vector space. Therefore, it cannot be a vector space.

- (b) A nonempty set that is closed under addition and scalar multiplication is a vector space. (×)

Explanation While being nonempty and closed under addition and scalar multiplication are necessary conditions for a vector space, they are not sufficient. By definition, a vector space must satisfy some axioms, that is, the commutativity and associativity of addition, the existence of an additive identity (zero vector), multiplicative identity and additive inverses and the distributive properties. Therefore, just being nonempty and closed under these operations does not guarantee that the set is a vector space.

You should not confuse the conditions for a set to be a **subspace** (which additionally must contain the zero vector) with the conditions for being a **vector space** (which also requires the existence of additive inverses and other axioms).

- (c) A plane in $\mathbb{R}^3$ is a two-dimensional subspace. (×)

Explanation A plane in $\mathbb{R}^3$ can be a two-dimensional subspace if it passes through the origin (the zero vector). However, if the plane does not pass through the origin, it is not a subspace. Therefore, not all planes in $\mathbb{R}^3$ are two-dimensional subspaces; only those that include the origin are.

- (d) Suppose U_1 and U_2 are subspaces of V . The intersection of U_1 and U_2 is a subspace of V . (✓)

Explanation The intersection of two subspaces U_1 and U_2 is indeed a subspace of V . This is because the intersection will contain the zero vector (since both U_1 and U_2 are subspaces), and it

will be closed under addition and scalar multiplication (since any linear combination of vectors in the intersection will also be in both U_1 and U_2). Therefore, the intersection of two subspaces is always a subspace.

- (e) $\{(x, y, z) \in \mathbb{C}^3 : x^3 = y^3\}$ is a subspace of $\mathbb{C}^3$. (×)

Explanation The set $\{(x, y, z) \in \mathbb{C}^3 : x^3 = y^3\}$ is not a subspace of $\mathbb{C}^3$. If we take two vectors in this set, say $(1, 1, 0)$ and $(1, \omega, 0)$ where ω is a cube root of unity (not equal to 1, that is, equal to either $e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ or $e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$). Both of these vectors satisfy the condition $x^3 = y^3$ (since $1^3 = 1^3$ and $1^3 = \omega^3$), but their sum $(2, 1 + \omega, 0)$ does not satisfy the condition because $2^3 \neq (1 + \omega)^3 = -1$. Therefore, this set is not closed under addition and hence cannot be a subspace.

- (f) A single vector by itself is linearly dependent. (×)

Explanation A set containing only the zero vector is linearly dependent. Thus, A single vector by itself is linearly independent **if it is not the zero vector**.

- (g) If $H = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is a basis for H . (×)

Explanation For $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ to be a basis for H , it must not only span H **but also be linearly independent**.

- (h) A basis is a linearly independent set that is as large as possible. (✓)

Explanation A basis of a vector space is defined as a linearly independent set of vectors that spans the entire vector space. This means that you cannot add any more vectors to the basis without losing linear independence, which is why it is considered "as large as possible" in terms of linear independence.

- (i) If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ are linearly independent lists of vectors in V , then $\{\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_m + \mathbf{w}_m\}$ is linearly independent. (×)

Explanation The linear independence of $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ does not guarantee that the set $\{\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_m + \mathbf{w}_m\}$ will also be linearly independent. For example, if $\mathbf{v}_i = -\mathbf{w}_i$ for all i , then $\mathbf{v}_i + \mathbf{w}_i = \mathbf{0}$ for all i , which would make the set $\{\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_m + \mathbf{w}_m\}$ linearly dependent.

- (j) V is a nonzero finite-dimensional vector space, and the vectors listed belong to V .

- i. If there exists a set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ that spans V , then $\dim V \leq p$. (✓)

Explanation If there exists a set of p vectors that spans V , then the dimension of V cannot be greater than p , because the dimension is defined as the number of vectors in a basis, and a spanning set can be reduced to a basis by removing linearly dependent vectors (removing nothing if it is already a basis). Therefore, $\dim V \leq p$.

- ii. If there exists a linearly independent set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V , then $\dim V \geq p$. (✓)

Explanation If there exists a set of p linearly independent vectors in V , then the dimension of V cannot be less than p , because the dimension is defined as the number of vectors in a basis, and a linearly independent set can be extended to a basis by adding more vectors (adding nothing if it is already a basis). Therefore, $\dim V \geq p$.

- iii. If $\dim V = p$, then there exists a spanning set of $p + 1$ vectors in V . (✓)

Explanation If $\dim V = p$, then there exists a basis of p vectors that spans V . By adding any vector from V that is not in the span of the basis, we can create a spanning set of $p + 1$ vectors. Therefore, there exists a spanning set of $p + 1$ vectors in V .

- iv. If there exists a linearly dependent set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V , then $\dim V \leq p$. (✗)

Explanation The existence of a linearly dependent set of p vectors does not necessarily imply that the dimension of V is less than or equal to p . For any vector space V , you can always find a linearly dependent set of p vectors by taking zero vectors or by repeating vectors, regardless of the actual dimension of V . Therefore, this statement is false.

- v. If every set of p elements in V fails to span V , then $\dim V > p$. (✓)

Explanation If no set of p vectors can span V , then no set of p vectors can be reduced to a basis of V . This means that the length of any basis of V must be greater than p , which implies that $\dim V > p$.

- vi. If $p \geq 2$ and $\dim V = p$, then every set of $p - 1$ nonzero vectors is linearly independent. (✗)

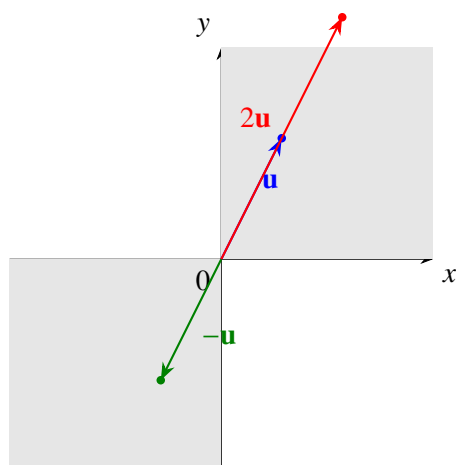
Explanation If $\dim V = p$, then there exists a basis of p vectors. However, not every set of $p - 1$ nonzero vectors will be linearly independent. For example, if you take $p - 2$ (if $p > 2$) vectors from the basis and add one more vector that is a linear combination of those $p - 2$ vectors, you will have a set of $p - 1$ nonzero vectors that is linearly dependent. Therefore, this statement is false. By the way, if $p = 2$, then every set of $p - 1 = 1$ nonzero vector is indeed linearly independent, but the statement is not true for all $p \geq 2$.

2. Let W be the union of the first and third quadrants in the xy -plane. That is, let $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy \geq 0, x, y \in \mathbb{R} \right\}$.

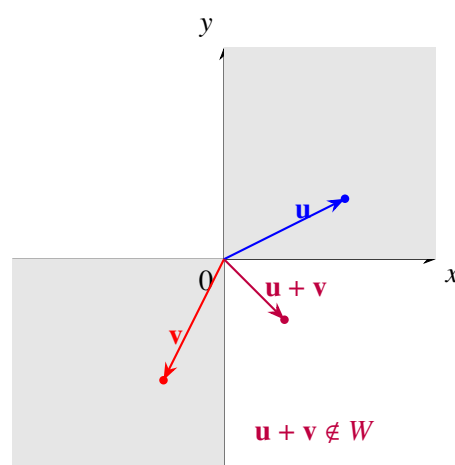
(a) If $\mathbf{u}$ is in W and c is any scalar, is $c\mathbf{u}$ in W ? Why?

(b) If $\mathbf{u}, \mathbf{v} \in W$, is $\mathbf{u} + \mathbf{v}$ in W ? Why?

(c) Is W a vector space?



(a) 标量乘法封闭: $c\mathbf{u} \in W$



(b) 加法不封闭: $\mathbf{u} + \mathbf{v} \notin W$

Solution:

(a) Yes. $c\mathbf{u} = [cx \quad cy]^T$. Since $\mathbf{u} = [x \quad y]^T \in W$, $xy \geq 0$. Thus $(cx)(cy) = c^2xy \geq 0$. Therefore $c\mathbf{u} \in W$. (Generally, the scalar field is $\mathbb{R}$ when working with real vector spaces.)

(b) No. Counterexample: $\mathbf{u} = [2 \quad 1]^T \in W$, $\mathbf{v} = [-1 \quad -2]^T \in W$, but $\mathbf{u} + \mathbf{v} = [1 \quad -1]^T \notin W$.

(c) No, because the addition on W is not closed.

3. Let X and Y be subspaces of a vector space V .

(a) Let $\mathbf{v} \in V$, $\mathbf{v} \notin X$. Prove that if $\mathbf{x} \in X$ then $\mathbf{x} + \mathbf{v} \notin X$.

(b) Use (a) to show that $X \cup Y$ is a subspace if and only if $X \subset Y$ or $Y \subset X$.

(c) Give an example that X and Y are both subspaces of a vector space V , but $X \cup Y$ is not.

Note: The union of subspaces is usually not a subspace, which is the reason for us to consider the sum of subspaces.

Solution:

(a) Suppose that $\mathbf{v} \in V$, $\mathbf{v} \notin X$, and $\mathbf{x} \in X$. If $\mathbf{x} + \mathbf{v} \in X$, then since X is a subspace, $\mathbf{v} = (\mathbf{x} + \mathbf{v}) - \mathbf{x} \in X$, which is a contradiction. Therefore, $\mathbf{x} + \mathbf{v} \notin X$. $\square$

(b) If $X \subset Y$ or $Y \subset X$, then $X \cup Y = X$ ($Y \subset X$) or $X \cup Y = Y$ ($X \subset Y$). Since X and Y are subspaces, $X \cup Y$ is also a subspace.

For the converse, it will suffice to show that if $X \not\subset Y$ and $Y \not\subset X$, then $X \cup Y$ is not a subspace. Since $X \not\subset Y$ and $Y \not\subset X$, there exist $\mathbf{x}, \mathbf{y} \in X \cup Y$ such that $\mathbf{y} \in Y$, $\mathbf{y} \notin X$ and $\mathbf{x} \in X$, $\mathbf{x} \notin Y$. Substitute $\mathbf{x}$ and $\mathbf{y}$ into $\mathbf{x}$ and $\mathbf{v}$ in (a), we know that $\mathbf{x} + \mathbf{y} \notin X$; interchange the role of $\mathbf{x}$ and $\mathbf{y}$ we get that $\mathbf{x} + \mathbf{y} \notin Y$. Hence $X \cup Y$ is not closed under addition, and it is not a subspace. $\square$

(c) Consider two subspaces of $\mathbb{R}^3$:

$$X = \{(x, 0, 0), x \in \mathbb{R}\}, Y = \{(0, y, 0), y \in \mathbb{R}\}.$$

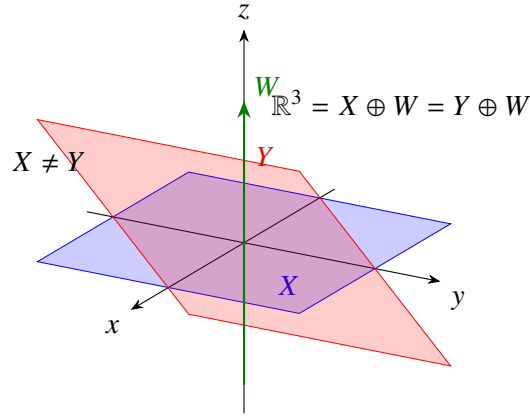
X includes all vectors parallel to x -axis while Y includes all vectors parallel to y -axis. Their union includes all vectors parallel to either x -axis or y -axis. However, consider the following two vectors in $X \cup Y$:

$$\mathbf{x} = (1, 0, 0), \quad \mathbf{y} = (0, 1, 0),$$

add them together and it turns out $\mathbf{x} + \mathbf{y} = (1, 1, 0)$, which is parallel to neither the x -axis nor the y -axis, and thus is not in $X \cup Y$. (举例需要满足: X 和 Y 确实是子空间; $X \cup Y$ 确实不是子空间。比如, 如果举上一题作为例子就是错的, 因为上一题的用于求并集的集合不是子空间)

4. Prove or give a counterexample: If X, Y, W are subspaces of V such that $X \oplus W = Y \oplus W$, then $X = Y$.

Solution: Counterexample: $X = \text{span}\{(1, 0, 0), (0, 1, 0)\}$, $Y = \text{span}\{(1, 0, 1), (0, 1, 0)\}$, and $W = \text{span}\{(0, 0, 1)\}$. $X \oplus W = Y \oplus W = \mathbb{R}^3$, but $X \neq Y$.



5. Suppose that V is a finite dimensional vector space. Let a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in V$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1} \in V$ such that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent. (Hint: Take for $\mathbf{v}_{r+1}$ any vector that cannot be represented as a linear combination $\sum_{k=1}^r \alpha_k \mathbf{v}_k$ and show that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.)

Proof:

Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are linearly independent but do not generate V . Then there exists a vector $\mathbf{v}_{r+1} \in V$ that cannot be expressed as a linear combination $\sum_{k=1}^r \alpha_k \mathbf{v}_k$. Choose such a $\mathbf{v}_{r+1}$. We show that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

Assume that $\alpha_1, \dots, \alpha_r, \alpha_{r+1}$ are scalars such that

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r + \alpha_{r+1} \mathbf{v}_{r+1} = \mathbf{0}.$$

If $\alpha_{r+1} = 0$, then $\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r = \mathbf{0}$, and by linear independence of $\mathbf{v}_1, \dots, \mathbf{v}_r$ we obtain $\alpha_1 = \dots = \alpha_r = 0$. If $\alpha_{r+1} \neq 0$, then we can solve

$$\mathbf{v}_{r+1} = -\frac{\alpha_1}{\alpha_{r+1}} \mathbf{v}_1 - \dots - \frac{\alpha_r}{\alpha_{r+1}} \mathbf{v}_r,$$

which contradicts the choice of $\mathbf{v}_{r+1}$. Therefore all coefficients are zero, and the vectors are linearly independent. $\square$

6. Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5)\}$, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

- Give the standard basis of $\mathcal{P}_4(\mathbb{F})$ and verify that it is indeed a basis. What is the dimension of $\mathcal{P}_4(\mathbb{F})$? What is the dimension of $\mathbb{F}^5$? (Hint: Can you see the relationship between them?)
- Find a basis of U . (Hint: Try to answer: Is $1 \in U$? Is $(x-2)(x-5) \in U$? Are there any other “independent” vectors? What is the possible size of the linearly independent set?)
- Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$. (Hint: The proof of Theorem 2.6.3 might help.)

Solution:

(1) The standard basis of $\mathcal{P}_4(\mathbb{F})$ is

$$\{1, x, x^2, x^3, x^4\}.$$

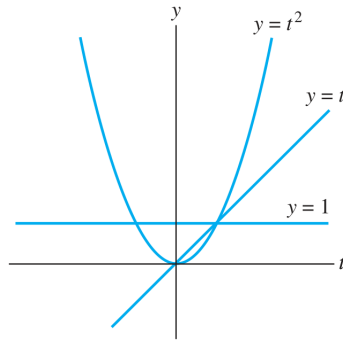


图 1: The standard basis for $\mathcal{P}_2(\mathbb{R})$.

Linear independence: If $a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 = 0$ (the zero polynomial), then all coefficients $a_i = 0$ because two polynomials are equal if and only if their coefficients coincide.

Spanning: Every $p(x) \in \mathcal{P}_4(\mathbb{F})$ can be written as $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$, so the set spans $\mathcal{P}_4(\mathbb{F})$.

Hence $\dim \mathcal{P}_4(\mathbb{F}) = 5$. The dimension of $\mathbb{F}^5$ is also 5.

Stack the coefficients of $p \in \mathcal{P}_4(\mathbb{F})$ into a column and we get a vector in $\mathbb{F}^5$; regard the scalars of a vector in $\mathbb{F}^5$ as the coefficients a polynomial and we get a $p \in \mathcal{P}_4(\mathbb{F})$. We say that the two spaces are **isomorphic** (同构的) via the coefficient map $(a_0, \dots, a_4) \leftrightarrow a_0 + a_1x + \dots + a_4x^4$. (You will learn this concept in the next chapter.)

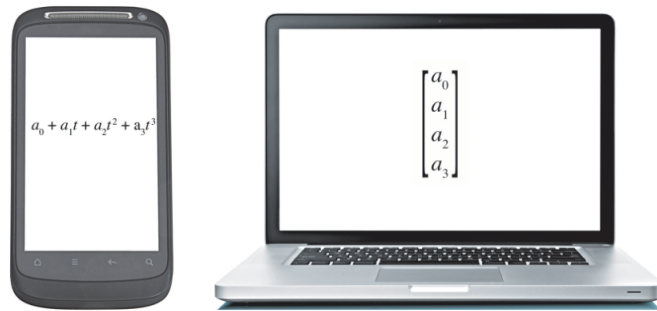


图 2: The space $\mathcal{P}_3(\mathbb{F})$ is isomorphic to $\mathbb{F}^4$.

(2) **Solution (1):** Suppose that $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$. The condition $p(2) = p(5)$ implies that

$$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = a_0 + 5a_1 + 25a_2 + 125a_3 + 625a_4 \implies a_1 = -7a_2 - 39a_3 - 203a_4$$

It's a single linear equation constraint in $\mathbb{F}^5$, which implies $\dim U = 4$.

Substitute the above equation into the expression of $p(x)$ and we get

$$p(x) = a_0 + a_2(x^2 - 7x) + a_3(x^3 - 39x) + a_4(x^4 - 203x)$$

which implies

$$\{1, x^2 - 7x, x^3 - 39x, x^4 - 203x\}$$

spans U . These four polynomials have degrees 0, 2, 3, 4 respectively; distinct degrees imply linear independence. Therefore we find a basis of U .

Solution (2): We look for linearly independent polynomials satisfying $p(2) = p(5)$:

- The constant polynomial 1 satisfies $1(2) = 1(5) = 1$, so $1 \in U$.
- $q(x) = (x - 2)(x - 5)$ satisfies $q(2) = q(5) = 0$, hence $q \in U$.
- $xq(x) = x(x - 2)(x - 5)$ and $x^2q(x) = x^2(x - 2)(x - 5)$ also vanish at 2 and 5, so they belong to U .

These four polynomials have degrees 0, 2, 3, 4 respectively; distinct degrees imply linear independence, which further implies that $\dim U \geq 4$.

Since $U \neq \mathcal{P}_4(\mathbb{F})$, $\dim U < \dim \mathcal{P}_4(\mathbb{F})$. Thus $\dim U = 4$. Therefore a basis of U :

$$\{1, (x - 2)(x - 5), x(x - 2)(x - 5), x^2(x - 2)(x - 5)\}.$$

(很多同学没有说明这些向量为什么是线性无关的, 或是没有说明为什么 4 个线性无关的向量就可以张成 U , 或是没有说明为什么 $\dim U = 4$; 维数、线性无关和张成这三个条件至少要正确说明两个)

(3) Since $\dim \mathcal{P}_4(\mathbb{F}) = 5$ and $\dim U = 4$, we need a one-dimensional subspace W such that $W \cap U = \{0\}$. Take for instance $W = \text{span}\{x\}$.

Method (1): Add x to the basis of U and we get a list

$$\{1, x, (x - 2)(x - 5), x(x - 2)(x - 5), x^2(x - 2)(x - 5)\}.$$

These polynomials have degrees 0, 1, 2, 3, 4 respectively; distinct degrees imply linear independence. Linearly independent list of the right length (dimension) is a basis, and thus we complete the basis of U to a basis of the whole space $\mathcal{P}_4(\mathbb{F})$. From Theorem 2.6.3, we get that W is the subspace such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Method (2): Because $x(2) = 2 \neq 5 = x(5)$, we have $x \notin U$; if $cx \in U$ with $c \neq 0$, then $x \in U$, which is a contradiction. Hence $W \cap U = \{0\}$. Dimensions add ($4 + 1 = 5$), so $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Thus one possible subspace is

$$W = \{ax : a \in \mathbb{F}\}.$$

(Other choices, e.g. $W = \text{span}\{x^4\}$, work as well.)

(维数、交集为 $\{0\}$ 、 $U + W = \mathcal{P}_4(\mathbb{F})$ 至少要出现其中两个)